



Why do we need this GenC learning Program?

Gen C learning program engages young talents with a comprehensive learning pathway, giving the millennials an opportunity to interact with Subject Matter Experts (SME), understand the corporate environment, and groom themselves.

Cognizant emphasizes on Learner Autonomy where students take charge of their own learning, with the available tools and resources. More focus is on “learning” than “teaching”. Get ready to embark your own learning adventure!

Program at a glance

Learning consisting of 3 Stages:

- Stage 1: Python Foundations for Automation, Python and SQL Assessment
- Stage 2: ETL Testing & Agile testing Practices, Interim Project Evaluation + Technical Evaluation
- Stage 3: Automation with Selenium & Integration, Modern AI-Assisted Automation(Common for all S
- Final Project Evaluation + Final Technical Evaluation

Program Highlights

- The complete learning journey is formalized using adult learning principles, where problem solving and applying the skills gained are given more importance than conceptual learning.
- Learner Autonomy is encouraged via Flipped Classroom, where the learning platform offers world class learning resources, and students would not be constrained by tutelage of an instructor.
- Get mentored by SME, whose motivation and guidance will help you accelerate in the learning journey.
- This program is applicable to Interns as well as GEN Cs.

Know Your Service Line – QEA (Quality Engineer and Assurance)

Cognizant Quality Engineering & Assurance (QEA) focuses on ensuring the quality and reliability of enterprise processes, applications, and systems. QEA offers a comprehensive range of services including intelligent and automated quality assurance, modernization assurance, and experience assurance. These services are designed to accelerate business and technology changes, improve customer experiences, and ensure regulatory compliance. QEA also emphasizes the importance of automation and AI to enhance testing efficiency and deliver high-quality outcomes. By leveraging these advanced technologies, QEA helps businesses achieve digital success and maintain competitive advantage in the evolving digital landscape.

How QEA Transforms Quality Assurance at Cognizant?

Cognizant's Quality Engineering & Assurance (QEA) significantly transforms quality assurance by accelerating digital transformation and ensuring robust, reliable applications and systems. This seamless transition minimizes disruptions and speeds up technology adoption.

By embedding quality at every development stage, QEA enhances customer satisfaction and loyalty through high-performance, user-friendly products. Automation and AI-driven quality assurance streamline processes, reducing manual efforts and speeding up release cycles, which boosts operational efficiency and cuts costs. Rigorous testing and validation ensure compliance with industry regulations, crucial for sectors like healthcare and finance.

QEA also supports innovation by providing a reliable foundation for launching new products and entering new markets. Additionally, automated testing frees employees to focus on strategic tasks, enhancing productivity and job satisfaction. These combined impacts help businesses achieve strategic goals, maintain a competitive edge, and deliver exceptional value to their customers.

Notable & Successful Stories on Quality Assurance by QEA Cognizant Team...

Cognizant's Quality Engineering & Assurance (QEA) team partnered with a major health plan provider to overhaul their software quality assurance processes, addressing several critical challenges. The existing methods were outdated and inefficient, struggling to keep pace with the healthcare industry's evolving needs.

Ensuring compliance with stringent healthcare regulations was a significant concern, and the manual, fragmented processes led to delays and increased operational costs. Additionally, there was a pressing need to improve software quality to enhance patient care and overall service delivery.

QEA implemented automated testing frameworks and AI-driven quality assurance tools, significantly reducing manual efforts and accelerating the testing process. Rigorous testing and validation processes were established to ensure compliance with healthcare regulations, including continuous monitoring and real-time compliance checks.

By streamlining quality assurance processes and integrating advanced technologies, QEA improved the overall efficiency of the software development lifecycle.

The focus on quality from the start ensured that end products were reliable and user-friendly, leading to better patient care and satisfaction. As a result, the health plan provider saw a significant improvement in software quality and compliance, which was crucial for meeting healthcare regulations.

The enhancements in software quality directly contributed to better patient care and service delivery, while the automation and streamlined processes led to faster release cycles and reduced operational costs.

The time-to-market for software updates was significantly reduced, allowing the client to respond more quickly to market demands. This project exemplifies how Cognizant's QEA services can transform quality assurance processes, ensuring high-quality outcomes.

[Health plan rebuilds software QA with Cognizant Case Study | Cognizant](#)

Tips for Successfully Carrying Best Practices for Implementing and Maintaining Robust IT Infrastructure with QEA.

Understand the Basics: Start with a solid understanding of software development and testing fundamentals. Familiarize yourself with key concepts such as the software development lifecycle (SDLC), different types of testing (e.g., unit, integration, system, and acceptance testing), and the importance of quality assurance.

Learn Automation Tools: Gain proficiency in popular automation tools like Selenium, JUnit, and TestNG. Automation is crucial for efficient testing, and being skilled in these tools will enhance your ability to create reliable and repeatable tests.

Adopt the Page Object Model (POM): Use the Page Object Model design pattern to create maintainable and reusable code. This approach separates test logic from page elements, making your tests more organized and easier to manage.

Focus on Continuous Integration/Continuous Deployment (CI/CD): Understand and implement CI/CD practices. Tools like Jenkins help integrate testing into the development pipeline, ensuring that code changes are tested continuously and deployed seamlessly.

Embrace Agile and DevOps Practices: Agile and DevOps methodologies promote collaboration, speed, and quality. Learn how to work in these environments, focusing on continuous improvement and adaptability.

Prioritize Test Coverage: Develop comprehensive test plans that cover all aspects of the application, including functional, performance, security, and usability testing. Ensure that your tests are thorough and cover various scenarios.

Leverage AI and Machine Learning: Explore how AI and machine learning can enhance testing processes. These technologies can help predict potential issues, optimize test cases, and improve the overall effectiveness of quality assurance.

Implement Data-Driven Testing: Use data-driven testing to run the same test with different sets of data. This approach increases test coverage and ensures that your application works correctly with various inputs.

Maximize Browser Window: Ensure that your tests maximize the browser window to capture full-page screenshots and interact with all elements. This practice helps improve test accuracy and reliability.

Validate with Assertions: Use assertions to validate the outcomes of your tests. Assertions help ensure that the application behaves as expected and quickly identify any issues.

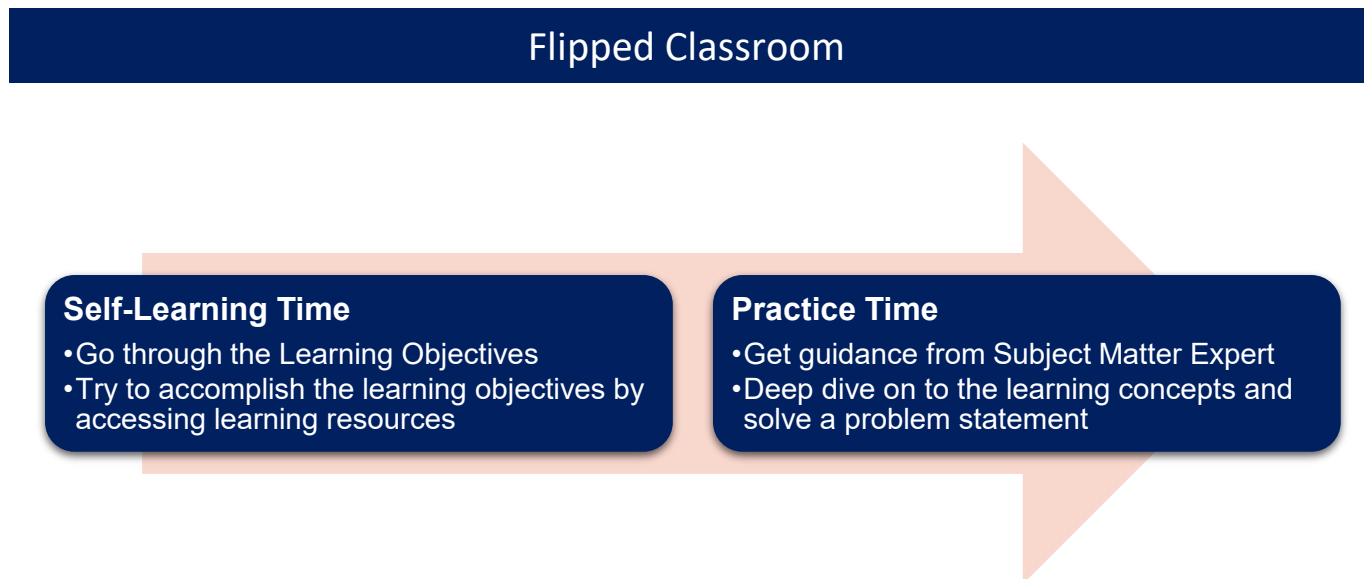
Continuous Learning and Improvement: Stay updated with the latest trends, tools, and best practices in quality assurance. Participate in training sessions, workshops, and online courses to continuously enhance your skills.

Collaborate and Communicate: Foster strong communication and collaboration with development, testing, and operations teams. Quality assurance is a shared responsibility, and effective teamwork is essential for success.

Learning Journey with Flipped Classroom

This program encourages you to be more autonomous learners during guided self-learning hours, completing the learning objectives on your own pace and style, and get ready for the hands-on practice time.

The complete learning path is set in the [GEN C Learn Platform](#), which you can login with SSO.



Recommended Program Sequence

The learning journey contains 3 stages, followed by a Business Aligned Project.

Stage	Course Segment	Course/Module	Day Wise
Stage 1: Python Foundations for Automation	Python Programming Refresher for Automation	Introduction to Python and its installation Python basics: variables, data types, control statements Python OOPs concepts, modules, and exception handling Working with text files and databases Advanced Concepts for Testing and ETL	Day 1- 5
	Data Structures Review	Focus on Dictionaries (for test data/config) and Lists/Tuples (for iterations/parameters). Key Concept: List Comprehensions and Dictionary Comprehensions for compact, readable data processing. Functions & Scope: Advanced understanding of function arguments (*args, **kwargs) and Lambda Functions (often used for quick sorts/filtering or in test helpers). Error Handling: Robust use of try...except...finally blocks, including catching specific exceptions (e.g., FileNotFoundError, TimeoutException in Selenium).	Day 6 - 7
	Advanced Python Concepts for Testing and ETL *Data Extraction *Data Transformation (Raw to Curated Layer) *Data Loading (Curated to Analytical)	Iterators and Generators: Understanding yield (critical for Pytest Fixtures and efficient data processing in ETL). Modules and Packages: Reinforcing how to structure larger projects (like a full automation framework) using __init__.py and proper import statements. Context Managers: Using the with statement (e.g., with open(...), with driver.switch_to.frame(...)). This ensures resources are properly handled and released—a cornerstone of reliable automation. Validation-Data Extraction (Source to Raw Layer): * Title: Record count Check Topics: * Title: Data Quality Check Topics: Data Transformation (Raw to Curated Layer): * Title: Duplicate Check Topics: * Title: Data Splitting Check Topics: * Title: Null Check Topics: Data Loading (Curated to Analytical) * Title: Data Integrity Check Topics: * Title: Load Source data to Table Topics:	Day 8 - 10
	Database Fundamentals with SQL Programming	Database Design SQL Basics with DML and DDL Statements Querying Database - Operators, Aggregate, String, Date Functions	Day 11 to 14
Stage 1 - Python and SQL Assessment			Day 15

Stage 2: ETL Testing & Agile testing Practices	Agile Functional Testing Integrated Project Deliverable: Agile and Functional Testing handson About QEA SL(Presentation)	Mastering Functional testing concepts, lifecycles, methodologies, and techniques. Hands-on project work: requirements analysis, test design, execution, and bug reporting. Exploring Agile/Scrum processes: sprints, user stories, planning, review, retrospective. Learning Jira tool for project, backlog, sprint, bug, and test case management. Mandatory hands-on projects, including automation concepts and creating test suites. Manual Software Testing fundamentals (STLC, V-Model, Test Types), practical application via a Live Project (FRS analysis, Test Case creation, Execution, and Bug Reporting), and essential skills in Agile/Scrum methodology and usage of the JIRA Tool for project, sprint, and defect management.	Day 11 - 17
		Trainer-Guided Integrated Project: Understanding QEA, Domain, Activity Readiness Trainer-Guided Integrated Project: Configured Jira Project & Backlog Management Trainer-Guided Integrated Project: Test Case Development in Jira (using Zephyr/similar plugin) Trainer-Guided Integrated Project: Test Execution & Status Reporting Artifacts Trainer-Guided Integrated Project: User story document Trainer-Guided Integrated Project: Retrospection Report Trainer-Guided Integrated Project: Sprint Backlog Placeholder to submit other Functional agile deliverables The Importance of QEA and Career Scope The Scope Debate: Quality Assurance vs. Software Development	
		Integrated Capability Test (ICT) - Functional Testing (Day 10)	
	Data Warehouse and Assurance with ETL Testing	Core phases (Extract, Transform, Load, Testing, Project): Introduction to ETL & Data Warehousing (DW) Data Sources & SQL Essentials (E) Advanced SQL for Transformation (E/T) The ETL Tool Landscape Data Extraction Techniques (E) Data Transformation & Cleansing (T) Data Loading Strategies (L) ETL Testing & Performance Optimization Real-time Streaming & Pipeline Development Capstone Project & Documentation	Day 18 to 28
Interim Evaluation			Day 29 and 30

Stage 3: Automation with Selenium & Integration	Automation & Selenium Overview	Understanding Software Testing and its types Challenges in Manual testing and how Automation overcomes them Functionality and components of Automation tools The role of Programming in Automation Overview, components, and advantages/disadvantages of Selenium WebDriver	Day 31 to 40
	Selenium Automation WebDriver, Assessment, Integrated Project, QEA Role specific Automation , Integrated Capability Test (ICT) Functional Testing	Introduction to WebDriver and its architecture Environment setup and configuration(Setup & configure WebDriver in Pycharm) Creating and executing automated test cases on multiple browsers Locators, commands, waits, browser and element commands Handling various web elements, alerts, iframes, and browser windows Advanced topics like handling date picker, web tables, mouse and keyboard actions, file uploads and downloads, screenshots, cookies, headless browser testing, and data-driven testing using MS-Excel Continuation of Integrated Project Deliverable: Agile and Functional Testing handson through Automation: Trainer-Guided Integrated Project: Automation Feasibility Report & Strategy Trainer-Guided Integrated Project: Configured Automation Environment & Project Structure Trainer-Guided Integrated Project: Automated Test Scripts (Individual Functional Scenarios) Trainer-Guided Integrated Project: Test Execution Reports and Status Trainer-Guided Integrated Project: Code Review and Feedback Session Artifacts	
	PyTest	Introduction and installation of PyTest Naming conventions, test cases, and fixtures Running tests in various ways, skipping tests, grouping tests Parallel testing, parameterization, command line arguments HTML report generation Master automation testing with Pytest framework. Continuation of Integrated Project Deliverable: Agile and Functional Testing handson through Automation: • Trainer-Guided Integrated Project: Enhanced Automation Test Scripts • Trainer-Guided Integrated Project: Comprehensive Error Handling and Reporting • Trainer-Guided Integrated Project: Utility Class for Advanced Functions • Trainer-Guided Integrated Project: Code Review & Refactoring Artifacts	
	GenAI Code Assist (GitHub Copilot): 3 Courses	Prompt Engineering & GitHub Copilot Usecase and Handbook Comprehensive GenAI Hands-On for Automation Developers	Day 41 to 43
	Logs & Page Object Model	Generating logs Page Object Model pattern and Page Factory	
	Hybrid Framework with POM – Live Project	Understanding Automation Frameworks and their types Designing a Hybrid Driven Framework step by step Project implementation with page objects, utilities, configuration files, automation, data-driven, and database test cases PyTest HTML reports, logs, email notifications Execution through command line and bat files	Day 44
	Continuous Integration (Git, Github & Jenkins)	Introduction to Continuous Integration Workflow of Git & Github, Git commands Overview of Jenkins and Maven integration Running Selenium test scripts through Jenkins Continuation of Integrated Project Deliverable: Agile and Functional Testing handson through Automation • Trainer-Guided Integrated Project: The "Sprint Automation" Hybrid Framework Codebase • Trainer-Guided Integrated Project: Complete Automated Regression/Sprint Suite • Trainer-Guided Integrated Project: Enterprise-Level Reporting Suite (Extent & Allure) • Trainer-Guided Integrated Project: CI/CD Ready Execution Artifacts • Trainer-Guided Integrated Project: Finalized Agile Documentation & Handover Package	Day 45 to 47
	Test Automation Frameworks(Robot Framework)	Continuation of Integrated Project Deliverable: Agile and Functional Testing handson through Automation Trainer-Guided Integrated Project: Refactored Automation Framework (Page Object Model Implementation) • Trainer-Guided Integrated Project: Data Management Strategy & Implementation (Apache POI) • Trainer-Guided Integrated Project: Execution Confirmation and Custom Logging Artifacts • Trainer-Guided Integrated Project: Updated Agile Artifacts & Traceability (Living Documentation) • Trainer-Guided Integrated Project: Robust Test Data Management via Test Automation Framework • Trainer-Guided Integrated Project: Comprehensive Execution Control • Trainer-Guided Integrated Project: Advanced Reporting with Extent Reports & Listeners • Trainer-Guided Integrated Project: Implementation of Assertions (Hard and Soft)	Day 48 to 50
	BDD using Behave	BDD versus Traditional Development What is Behave? Install Behave and create Features Files and Step Definitions Using hooks Parameterizing the steps Data Driven Testing using Scenario Outline Tags Create Cucumber BDD testing framework using Python, Behave Read and write Cucumber BDD style tests in Python Cucumber BDD Framework supporting both Frontend with BDD Selenium and Backend with pure BDD Python Tips and tricks of finding elements, debugging issues, shortcuts with IDE, test methodologies, setup py file	Day 51
	Behave BDD Hybrid Framework with PageObjectModel	Creating Feature files and Step Definitions Implemented Selenium Python code for all Scenarios Implementing hooks and utilities Implementing Page Object Model Optimizing Page Object Model Implementing Data Driven Testing No need for reading data from excel files and instead getting data from Generating Allure Reports Git, GitHub and Jenkins	Day 52 to 53
STAGE: Modern AI-Assisted Automation	GenAI Code Assist (GitHub Copilot)	Comprehensive GenAI Hands-On for Automation Developers • Trainer-Guided Integrated Project: Chat Endpoint Basics • Trainer-Guided Integrated Project: Understanding Tokenization & Context Limits(Estimate & Optimize Tokens) • Trainer-Guided Integrated Project: Prompt Optimization via /ai/chat/with-usage • Trainer-Guided Integrated Project: Generate Edge Cases • Trainer-Guided Integrated Project: From User Story to Feature File (AI-Driven BDD Generation) • Trainer-Guided Integrated Project: Generate Locators for SauceDemo Login (Dynamic Locator Generation) • Trainer-Guided Integrated Project: AI-Augmented Framework • Trainer-Guided Integrated Project: Prompt Engineering Game	Day 54 to 55
			Day 56 to 57

Common for all Stages	Handson, CC,Business mentor connect, BH Training(Weekly 4 hours , total 40 hours)				
	Stage 1 & 2 - Experimental - Field Trips	Weekly Once live project visit will enable GenCs to get the feeling of real time work environment.			
		This trip includes Test Lab visit which is established for the customer/account to understand the set-up			
		Stage 2 - Final Integrated Project Evaluation + Final Technical Evaluation			
			Day 58 to 60	Evaluation	

Key Learning Components of the Program

Cognizant has collaborated with Udemy to provide world class learning videos for the evolving future of work. These Udemy programs are woven into a learning path, empowering you to plan and learn at your style.

The program also connects you with Subject Matter Experts (SMEs) to get the professional guidance on your queries in the learning journey.

The program doesn't ONLY concentrate on the technical skilling, but also on the shaping up of the Behavioral skills. **Behavioral learning** would be done in ILT mode, with few Self-paced learning modules too.

Evaluation Model

The program continuously evaluates if you can apply those self-learnt skills to solve a real-time business problem. Depicted below are the four key learning components, which are distributed across the learning journey for the purpose of continuous evaluation.

- Interim Evaluation (Project + Technical) through Video Interview
- Final Evaluation (Project + Technical) through Video Interview

The above evaluation components will attribute to the Performance Health Status (PHS) of a GenC. Additional Learning Components like Hands-On, Code Challenges and ICTs will help you to enhance your expertise level.

Interim & Final Evaluation Approach

Below is the Evaluation Structure for GenC Learning Journey

The interim evaluation will be held halfway through the learning journey, while the final evaluation will take place at the end of the learning journey.

During the interim evaluation, the GenC will be interviewed by a Technical Subject Matter Expert (SME) from the Business Unit (BU) to assess your knowledge through a technical discussion. Additionally, the Mini project completed by the GenC will also be evaluated by the BU SME. Please note that the Mini project is an individual activity and not a group activity.

During the final evaluation, the GenC will again be interviewed by a Technical SME from the BU to assess their knowledge through a technical discussion. In the same evaluation call, the Hackathon project completed by each group will be evaluated by the BU SME. Please note that the Hackathon project is a group activity and not an individual activity.

Preparatory Learning

For this automation track training, here are some preparatory learning materials that can be helpful for the interim and final technical evaluations. Preparatory learning material refers to resources, materials, and content that you can use to prepare. These materials are designed to provide you with the necessary knowledge, skills, and understanding to succeed in your learning objectives.

Remember, apart from these learning materials, actively participating in the training sessions, asking question and practicing hands-on exercises will greatly contribute to your overall learning and preparedness for the evaluations.

Program Completion Criteria

Stage 1 (Pre-requisites)			
Gating Criteria: Qualifier Assessment			
Stage 2 & Beyond (Advanced Skills)			
Gating Criteria: Performance Health Status is Green			
GenC/Intern Training	Evaluation Components	Pass Criteria	Evaluation Done by
Performance Health Status - PHS (only from Stage 2)	Interim Evaluation (Project + Technical)	Green, 1 Attempt	BU SME
	Final Evaluation (Project + Technical)	Green, 2 Attempts*	

Outcome of Interim / Final evaluation will be RED, AMBER or GREEN status

Note: 100% Completion of Hands On in Stage 1 is mandatory for qualifier assessment and 100% Completion of Hands On in Stage 2 is mandatory for interim / final evaluation eligibility.

Key Check Point Intervals in the Learning Journey

RAG as PHS (Performance Health Status)

The program continuously evaluates if you are able to apply those self-learnt skills to solve a real time business problem. Depicted below are the two key learning components, which are distributed across the learning journey for the purpose of continuous evaluation.

Interim Evaluation:

- During the interim evaluation, the GenC will undergo a video interview on the learning platform.
- This interview will be conducted by a Tech SME from the BU.
- The purpose of this evaluation is to assess the GenC's knowledge and understanding of the skills covered in the training program up to the halfway point.
- It also encompasses an evaluation of the GenC's progress in their Project case study.
- The evaluation will involve a technical discussion as well as an assessment of the Project progression to gauge the GenC's proficiency in the skills learned thus far.
- There will be ONLY one attempt.

Final Evaluation:

- For the final evaluation, the GenC will participate in a video interview conducted by a Tech SME from the BU.
- This evaluation aims to assess the GenC's knowledge and expertise in all the skills covered throughout the entire training program.
- Similar to the interim evaluation, this assessment will involve a technical discussion via a video interview on the learning platform, along with a project evaluation to assess the GenC's capabilities and their Project case study progress.
- It serves as a comprehensive evaluation of the GenC's skills and capabilities acquired during the training.
- In Final Evaluation , Other 1 attempt is given when student PHS is AMBER/RED. (Attempt 1 is not applicable if the student is **Red** in Interim and Final Evaluation - Attempt 1)

The above evaluation components will attribute to the Performance Health Status (PHS) of a GenC. Additional Learning Components like Hands-On, Quizzes, CCs, and ICTs will help you to enhance your expertise level.

Mandatory Hands-On Exercise Completion :

- Completion of 100% of the hands-on exercises is mandatory to qualify for the stage 1 assessment, interim, and final evaluations.

Table - Check Point Intervals

Check points	Interpreting Status
Interim Evaluation	- Green - On Track for Graduation - Red /Amber - There will not be any re-attempts given
Final Evaluation	- Green - On Track for Graduation - Red /Amber – Only 2 attempts are given - (Attempt 2 is not applicable if the student is Red in Interim and Final Evaluation- Attempt 1) Note: If student fails after the applicable re-attempts, they will be considered as “Not-Graduated”.



Icebreaker session will be conducted for a duration of initial **5 days**. During the session, various topics related to Corporate Induction, Talent Management, Cognizant Agenda on Core Values, Leader Talks, Alumni, BU Mentor connects will be covered. Followed by icebreaker, technical training will kick start.

Following sessions will be covered during the **5 days** of icebreaker

- Corporate Induction
- Talent Manager Connect
- Cognizant Agenda Sessions on Core Values
- Leader Talks (Academy) and many more...

A recommended day-wise schedule is provided below for the learning, with the learning content for the day, the practice hands-on and extended hands-on to be done for the day or

any other activities are listed. Few days might be interleaved to accommodate the extension due to Behavioral Training.

Mock Interview Session: This session is conducted to test candidate understanding and enhance candidates' best preparation for Interim & Final Evaluations.

Software Requirements

Below are the software's need to be installed in system from Company Portal, before start proceeding with selenium Handson and projects like mini project and hackathon.

Category	Software/Tool	Latest Version	Notes
Programming	Python	3.14.2 (Stable) or 3.13.11 (LTS)	The core language for scripting and automation.

Development Env.	PyCharm	2025.3	The IDE recommended for the course, which is now a unified product (Community features are free).
Database	SQL DBMS (e.g., MySQL)	Varies	A specific database system will be needed to practice SQL fundamentals.
Test Automation	Selenium WebDriver	4.38.0	The primary web automation framework.
Testing Framework	PyTest	Varies	A Python testing framework; installed via <code>pip</code> .
Project Mgmt.	Jira Software (Data Center)	11.3.0 (LTS is 10.3.14)	Free version is sufficient. Used for agile practices, bug tracking, and test case management.
CI/Version Control	Git/GitHub	Varies	Git is a local VCS, while GitHub is a web platform for hosting repositories.
CI/CD	Jenkins	Varies (LTS release line)	An automation server for Continuous Integration.
AI Assistance	GitHub Copilot	Varies (Service updates)	An AI code assistant that integrates with the IDE.
Data Handling	MS-Excel	Varies	Used for data-driven testing scenarios

NOTE: If genc's are facing any challenges while installation, trainer/mentor will help to install the software.

Objective to use the software:

- Developing the project
- Writing the code
- Debugging the errors
- Testing the applications

Schedule – Stage 1: Day 1 to 5

Day 1 to 5 will be focusing on Python Programming Refresher for Automation

Udemy learnings are recommended in the Platform to understand the fundamental concepts. Apply the concepts learned and solve the Hands-on and Practice Case studies as recommended below.

Continuous Learning: Technical Hands-on through Tekstac

Post Completion of the ILT session, Trainee must Practice the Handson provided in the Tekstac Platform . In build code editor is available in this platform and Handson are automatically evaluated when the code is compiled in the editor.

BU Mandatory - Self Learning Contents – Python Programming Refresher for Automation

Day 1 - 5

Python Programming Refresher for Automation

Continuous Learning: ILT Enablement

Course Segment	Course/Module
Python Programming Refresher for Automation	Introduction to Python and its installation Python basics: variables, data types, control statements Python OOPs concepts, modules, and exception handling Working with text files and databases Advanced Concepts for Testing and ETL

Continuous Learning: Technical Enablement through Self-learning

Learn and Practice



[Learn Selenium with Python, PyTest & Frameworks.](#)

Refer section 2 in this Udemy course and complete the corresponding learnings.

Section 2: Java Programming

Continuous Learning: Technical Hands-on through Tekstac Mandatory Hands-on

- Welcome Message
- Virtual Assistant
- Hospital Billing
- Stationary Shop
- Theme Park Pricing
- Cash Exchange
- Non-Working Doctors
- Product Sales
- Participant Registration and Preference Analysis
- Student Electives
- Sort the Student Details Dictionary

- House Rent Calculator
- Airline Reservation System
- Voting System
- Fitness Tracker Analyzer
- Generate Customer Id
- Online Shopping Cart System
- Pattern Generator
- Basket Ball Player Analysis
- Percentage of marks - Lambda Functions
- Employee Appraisal
- Product Information
- Filter the Countries Sachin has Played Against - CSV file
- Filter Customers - JSON File
- Classic Curve Members
- Complex Addition
- Area calculation - Method Overriding
- Tournament - Method overloading
- Vehicle Premium
- Password Update
- Data Encryption
- Lucky Number - Generator
- Project Allocation - Iterator
- Summer Camp - Decorator

[Additional Learnings](#)

[Python Beginners Primer V1](#)

[Handson](#)

[Real-Time Hands-On for Python Programming Module](#)

Please click this link to download Handson related topics to be learnt as part of this milestone. Once practiced Handson , please upload the solution file and upload here for evaluation by trainer.

Schedule – Stage 1: Day 6 to 7

Day 6 to 7

Data Structures Review

Day 6 to 7 will be focusing on Data Structures Review

Udemy learnings are recommended in the Platform to understand the fundamental concepts. Apply the concepts learned and solve the Hands-on and Practice Case studies as recommended below.

Continuous Learning: Technical Enablement



[Python Data Structures & Algorithms + LEETCODE Exercises](#)

Refer to all the sections of the Udemy course

On Day 6 - Once completing Udemy course, Learnings, all Hands-on & Practice assessments related to Python programming skill appear for the **Python - Code Challenge Assessment**. This will help you to assess and apply the concepts of the skill learnt in the platform.

Assess-Type-1: Code Challenge – Python

Assessment Duration: 2 hrs.

Assessment Attempts: 3

No. of Questions appear in each attempt: 1

Day 8 to 10

Advanced Python Concepts for Testing and ETL

Continuous Learning: Technical Enablement

Advanced Python Concepts for Testing and ETL *Data Extraction *Data Transformation (Raw to Curated Layer) *Data Loading (Curated to Analytical)	Iterators and Generators: Understanding yield (critical for Pytest Fixtures and efficient data processing in ETL). Modules and Packages: Reinforcing how to structure larger projects (like a full automation framework) using __init__.py and proper import statements. Context Managers: Using the with statement (e.g., with open(...), with driver.switch_to.frame(...)). This ensures resources are properly handled and released—a cornerstone of reliable automation. Validation-Data Extraction (Source to Raw Layer): * Title: Record count Check Topics: * Title: Data Quality Check Topics: Data Transformation (Raw to Curated Layer): * Title: Duplicate Check Topics: * Title: Data Splitting Check Topics: * Title: Null Check Topics: Data Loading (Curated to Analytical) * Title: Data Integrity Check Topics: * Title: Load Source data to Table Topics:	Day 8 - 10
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Python for ETL Testing:

Title: Know How & Installation of Python Topics:

- 1. Installation & Setup
- 2. Variables and Printing
- 3. Numbers in Python -- If time permits
 - a. Add
 - b. Subtract

- c. Multiply
- d. Divide
- i. Floor
- ii. Ceiling
- e. Modulo

Title: Lets Play with Numbers & Strings Topics:

- 4. Strings
- a. Formatting
- b. Concatenate
- c. Substring
- 5. User Input
- Activity - Start Building a Calculator with proper messages

Title: Activity - Building a Calculator Topics:

- 6. Booleans
- 7. If loop
- 8. And / or
- Activity - Complete Building a Calculator

Title: Data Structures of Python - 1 Topics:

- Data Structures of Python
- 9. Lists
- 10. Tuples
- Along with its operations

Title: Data Structures of Python - 2 Topics:

- 11. Dictionaries
- 12. Sets
- Along with operations

Title: Loopy Whoopy - For & While Loop Topics:

- 14. For loop
- 15. While loop

Title: Break the loops Topics:

- 16. Destructuring
- 17. Break and Continue
- 18. Else in loop

Title: List & Loops together Topics:

- 19. List Comprehensions
- 20. List Slicing
- 21. Comprehensions with conditions

Title: Activity - Json Handling Topics:

- 22. Zip n enumerate functions
- Activity - Movie Details Display -JSON Handling

Title: Creating Functions Topics:

- 23. Functions -
- a. Arguments & Parameters
- b. Execution

Title: Lambda Functions Topics:

- 24. Lamda Functions

Title: Error Handling Topics:

- Try Except Blocks

Title: Pandas Library - 1 Topics:

- Pandas library -
- a. Read csv, excel
- b. Print Columns, Head, Size
- c. Select Columns to print
- d. Select Rows to print

Title: Pandas Library - 2 Topics:

- Pandas library
- a. Write to csv, excel
- b. Operations on Dataframe - Drop Na, Drop Duplicates, Copy Dataframes

Title: Activity - Pandas Library Topics:

- Data Compare & Operations

Title: Introduction to key Python Libraries Topics:

- Flask
- Tkinter, pySimpleGUI
- Selenium
- Pyodbc
- etc

Title: Q&A and Closure Topics:

- Q &A
- Discuss internal/external advance courses and certifications

Python for ETL Testing- Bootcamp:

Validation-Data Extraction (Source to Raw Layer)

Title: Record count Check Topics:

If record count in source > 0, then record count in target should be > 0

Title: Data Accuracy Check Topics:

Basic field level value check for one record in the source Vs target

Title: Data Quality Check Topics:

Checking data in target table as per data type. E.g. An Integer column should have an Integer value

Data Transformation (Raw to Curated Layer)

Title: Duplicate Check Topics:

There should not be any duplicate records in the target as per key(s)

Title: Data Mapping Check Topics:

Data check as per mapping E.g. Full Name column in target table should be in UPPER CASE e.g. JOHN DOE and should match the value from source

Title: Data Splitting Check Topics:

Validate Street, City, Zip Code and Country column are correctly populated from the one source Column Address. E.g. 100 Cold Spring, Charter Hill, USA, 23472

Title: Derived Column Check Topics:

Paid Amount Column is calculated as (Allowed Amount - Patient Responsibility)

Title: Null Check Topics:

Mandatory fields should not be Null

Title: Data Consistency Check Topics:

Same type of data should not have different formats. E.g. Date column

Title: Incremental Data Check Topics:

Incremental load count should be greater than the target data count

Data Loading (Curated to Analytical)

Title: Data Integrity Check Topics:

Customer Information is available in parent and child tables

Title: S2T Check Topics:

Full Volume source to Target check for one table

Title: Load Source data to Table Topics:

Load data from Excel & Text files to a target table

Python for ETL Testing Handson:

- Problem1
- Problem2
- Problem3

Data Transformation (Raw to Curated Layer)

Data Transformation

Data Loading (Curated to Analytical)

Data Loading

Schedule – Stage 1: Day 11 to 14

Day 11 to 14

Database Fundamentals with SQL Programming

Udemy learnings are recommended in the Platform to understand the fundamental concepts. Apply the concepts learned and solve the Hands-on and Practice Case studies as recommended below.

Continuous Learning: Technical Enablement

Learn the basics of SQL



[Relational Database Design](#)

Refer to all the sections of the Udemy course

Go through web pages for learning below specific topic- **No SQL overview**

[Introduction to NoSQL](#)



Refer sections 1 to 12, and 14 in this Udemy course and complete the corresponding learnings. Go through web pages for learning below specific topics

[RANK Function](#)

Continuous Learning: Technical Hands-on Mandatory Hands-on

- Create Borrowing Table
- Alter Table User
- Alter Table Book
- Remove Table
- Insert User Table
- Update Record
- Delete Record
- Diverse Selection
- Email Providers
- Book Event
- Upcoming Fair
- Routine Checks
- Security Measure
- May Birthday
- Age Demographics
- Author Detail

Mandatory Hands-on

- Price Analysis
- Updated Email
- Lended Books
- The Chronicles of Authors and Their Tales
- Date Format
- Exploring Users' Literary Journeys
- Active Borrowings with User and Book Details
- Same Authors
- User Borrowing Summary
- Maximum Fine Per User
- Top Users
- User Count by Title and Author

Additional Learning:

Technical Quizzes:

SQL basics with DML and DDL Statements

Querying Database - Operators, Aggregate, String, Date Functions

INTEGRATED ASSESSMENT

ANSI SQL - SKILL BASED ASSESSMENT

Please take this assessment as practice

On Day 13 - Once completed Udemy course, Learnings, all Hands-on & Practice assessments related to SQL skill appear for the **INTEGRATED ASSESSMENT**. This will help you to assess and apply the concepts of the skill learnt on the platform.

Assessment Duration: 2 hrs.

Assessment Attempts: 3

No. of Questions appear in each attempt: 1

MOCK Stage1 Assessment

On Day 14 - Once completing Udemy course, Learnings, all Hands-on & Practice assessments related to Python and SQL skill appear for the **Mock Stage1 Assessment**. This will help you to assess and apply the concepts of the skill learnt on the platform.

Assessment Duration: 3 hrs.

Assessment Attempts: 3

No. of Questions: 3 [Python -2 ; SQL -1]

Day 15

Once completing Udemy course, Learnings, all Hands-on & Practice assessments related to Python medium complexity, SQL medium complexity skill appear for the **Stage 1 Assessment**. This will help you to assess and apply the concepts of the skill learnt in the platform. Question Pattern - 2 SBA Python Questions and 1 SQL question approximately spend around 1 hr. for each SBA Question. Score acquiring in 1st instance will be getting into consideration.

Stage1 Assessment

Assessment Duration: 3 hrs.

No. of Questions: 3 [Python -2 ; SQL -1]

Schedule – Stage 2: Day 11 to 30

Udemy learnings are recommended in the Platform to understand the fundamental concepts. Apply the concepts learned and solve the Hands-on and Practice Case studies as recommended below.

Agile Functional Testing

Topic #	Topic Name	Topic Objective	References
Module 1	Agile Software Testing Concepts	Understanding SDLC and STLC	Theory and Foundations
		Distinguishing between Project and Product in testing context	
		Exploring V-Model and the roles of QA, QC, and QE	
		Differentiating between White Box and Black Box Testing	
		Delving into Static Testing and Dynamic Testing	
		Conceptualizing Verification and Validation	
		System Testing Types (GUI Testing)	
		System Testing Types (Functional vs. Non-Functional Testing)	
		Test Design Techniques and strategies	Includes Re-Testing, Regression Testing, Exploratory Testing, Adhoc, Sanity, Smoke Testing, and End-To-End Testing.
		Understanding STLC phases (Use case, Test scenario, Test case)	
		Understanding STLC phases (Test Environment, Execution, Defect Reporting)	
		Understanding STLC phases (Test Closure, and Test Metrics)	
Module 2	Agile Software Testing Project	Introduction to the Live Project	Practical Application
		Extracting Functional Requirements from FRS	Requirement Analysis
		Creating comprehensive Test Scenarios and Test Cases	Writing Test Artifacts
		Execution of Test Cases	
		Reporting and Tracking Bugs	Defect Management
		Test Sign-off	Completion Phase
Module 3	Agile/Scrum Process	Overview of Agile and Scrum methodologies	Agile Fundamentals
		Understanding Scrum Teams, Sprints, and User Stories	
		Estimating User Stories and defining 'Definition of Done' and 'Definition of Ready'	
		Exploring Sprint Activities (Planning, Refinement, Review, Retrospective)	
Module 4	JIRA Tool Implementation	Installation and Configuration of JIRA	Hands-on Tool Setup
		Project creation in Jira and user management	

		Creating Backlog, Epic, and User Stories in JIRA	Requirements in JIRA
		Sprint creation and management	
		Understanding the Sprint life cycle in JIRA	
		Bug creation in Jira	Defect Reporting in JIRA
		Writing test cases in JIRA	Test Case Management
		Creating Test Cycles and executing/updating Test Cases in Jira	Test Execution in JIRA

Day 16 - 22

Functional Testing

Continuous Learning: Technical Enablement

Learn the basics of Agile Fundamentals and Software Testing Life Cycle



[Learn Manual Software Testing + Agile with Jira Tool](#)

Learn all the sections in this Udemy course to gain deep understanding about the modules below

Please click this link to download fundamental Agile topics to be learnt as part of this milestone. Recommend trainer to cover advanced topics on your own

Module : Software Testing Concepts

- Understanding Software Development Life Cycle (SDLC) and Software Testing Life Cycle (STLC)
- Distinguishing between Project and Product in the context of testing
- Exploring V-Model and the roles of QA (Quality Assurance), QC (Quality Control), and QE (Quality Engineering)
- Differentiating between White Box and Black Box Testing
- Delving into Static Testing and Dynamic Testing
- Conceptualizing Verification and Validation in software testing
- System Testing Types, including GUI Testing and Functional vs. Non-Functional Testing
- Test Design Techniques and strategies like Re-Testing, Regression Testing, Exploratory Testing, Adhoc Testing, Sanity, Smoke Testing, and End-To-End Testing
- Understanding Software Testing Life Cycle (STLC) phases, including Use case, Test scenario, Test case, Test Environment, Execution, Defect Reporting, Test Closure, and Test Metrics.

Module : Software Testing Project

- Introduction to project
- Extracting Functional Requirements from Functional Requirement Specifications (FRS)
- Creating comprehensive Test Scenarios and Test Cases

- Execution of Test Cases
- Reporting and Tracking Bugs
- Test Sign-off

Phase : Agile Testing + Jira Tool

Agile/Scrum Process:

- Overview of Agile and Scrum methodologies
- Understanding Scrum Teams, Sprints, and User Stories
- Estimating User Stories and defining 'Definition of Done' and 'Definition of Ready'
- Exploring Sprint Activities such as Sprint Planning, Backlog Refinement, Sprint Review, and Sprint Retrospective

Jira Tool:

- Installation and Configuration of JIRA
- Project creation in Jira and user management
- Creating Backlog, Epic, and User Stories in JIRA
- Sprint creation and management
- Understanding the Sprint life cycle in JIRA
- Bug creation in Jira
- Writing test cases in JIRA with Zephyr plugin
- Creating Test Cycles and executing/updating Test Cases in Jira

Continuous Learning: Technical Hands-on

About QEA

Mandatory Hands-on

- The Importance of QEA and Career Scope
- The Scope Debate: Quality Assurance vs. Software Development

2. Mini Project

As part of interim evaluation, Trainees must do below integrated projects has been shared below and demonstrate these projects in the interim evaluation to the evaluator.

- Trainer-Guided Integrated Project: Activity Readiness
- Trainer-Guided Integrated Project: Configured Jira Project & Backlog Management
- Trainer-Guided Integrated Project: Test Case Development in Jira (using Zephyr/similar plugin)
- Trainer-Guided Integrated Project: Test Execution & Status Reporting Artifacts
- Trainer-Guided Integrated Project: User story document
- Trainer-Guided Integrated Project: Retrospection Report
- Trainer-Guided Integrated Project: Sprint Backlog
- Placeholder to submit other Functional agile deliverables

✓ **You should have introduced more than one sprints**

- ✓ Ensure you have in-sprint automation concepts introduced as part of project
- ✓ Ensure you have created Regression suite and smoke suite
- ✓ Ensure your regression suite consists of minimum 25 test cases and test scripts
- ✓ Ensure your smoke suite covers end to end business functionality

Additional Learning:

Technical Quizzes:

Functional Testing

On Day 22 , Once completing Udemey course, Learnings, all Handson & Practice assessments related to **Functional Testing** skill appear for the **Functional Testing -ICT Assessment**. This will help you to assess and apply the concepts of the skill learnt in the platform. Attempts allowed to practice the assessment is 3.

Assessment Duration: 4 hrs

Assessment Attempts: 3

No.of Questions appear in each attempt : 1

Functional Testing -Assess Type 2 Skill Based Assessment.

- Read the requirements carefully of the Assessment Question.
- Write the Defect Description for the Test Case in Excel Sheet
- Submit the Test deliverables

Schedule – Stage 2: Day 23 to 28

Day 23 to 28 will be focusing on Data Warehouse and Assurance with ETL Testing

Topic: Introduction to ETL and Data Warehousing

Description: Overview of ETL process, importance, and real-world use cases

Topic: Data Warehousing Concepts and Architecture

Description: Star & Snowflake Schema, Fact & Dimension Tables, Data Marts

Topic: Understanding Data Sources and Types

Description: Types of data (structured, semi-structured, unstructured), sources (DBs, APIs)

Topic: Introduction to ETL Tools (Informatica, Talend, SSIS)

Description: Overview of popular ETL tools, comparison of features

Topic: Data Extraction Techniques

Description: Extracting data from DBs, APIs, flat files, web scraping

Topic: Data Transformation Techniques

Description: Transforming data: cleansing, standardization, deduplication

Topic: Data Loading Strategies

Description: Loading strategies: batch vs real-time, bulk loading

Topic: ETL Hands-on: Simple Data Pipeline

Description: Implementing a simple ETL pipeline with a chosen tool

Topic: Error Handling and Data Quality

Description: Handling null values, duplicates, outliers, data integrity issues

Topic: ETL Performance Optimization

Description: Performance tuning, indexing, partitioning, parallel processing

Topic: Incremental Data Load vs Full Load

Description: Difference between incremental and full data loads, use cases

Topic: Real-time Data Processing (Streaming ETL)

Description: Introduction to Apache Kafka, Spark Streaming for real-time ETL

Topic: ETL Testing and Validation

Description: Writing and executing ETL test cases, debugging errors

Topic: ETL Project Planning and Dataset Selection

Description: Selecting a dataset, defining ETL workflow and architecture

Topic: Building ETL Pipelines - Hands-on

Description: Developing an end-to-end ETL pipeline with real data

Topic: Debugging and Troubleshooting ETL Jobs

Description: Identifying and fixing ETL job failures, logging mechanisms

Topic: Project Completion and Documentation

Description: Finalizing ETL workflow, documenting processes

Topic: Final Presentation

Description: Presenting project, assessment, and feedback

Evaluation: **Interim Project Evaluation + Technical Evaluation**

The overall Interim Technical evaluation will cover topics specified up to the Interim Evaluation in the curriculum and will be conducted via a video interview on the GenC Learn Platform.

There are no Additional Attempts for the Interim Evaluation. RAG Achieving in the first attempt is considered as final RAG for Interim Evaluation.

Schedule – Stage 3: Day 31 to 60

Stage 3: Automation with Selenium & Integration

Automation & Selenium Overview	Understanding Software Testing and its types Challenges in Manual testing and how Automation overcomes them Functionality and components of Automation tools The role of Programming in Automation Overview, components, and advantages/disadvantages of Selenium WebDriver	Day 31 to 40
Selenium Automation WebDriver, Assessment, Integrated Project, QEA Role specific Automation , Integrated Capability Test (ICT) Functional Testing	Introduction to WebDriver and its architecture Environment setup and configuration(Setup & configure WebDriver in Pycharm) Creating and executing automated test cases on multiple browsers Locators, commands, waits, browser and element commands Handling various web elements, alerts, iframes, and browser windows Advanced topics like handling date picker, web tables, mouse and keyboard actions, file uploads and downloads, screenshots, cookies, headless browser testing, and data-driven testing using MS-Excel Continuation of Integrated Project Deliverable: Agile and Functional Testing handson through Automation: Trainer-Guided Integrated Project: Automation Feasibility Report & Strategy Trainer-Guided Integrated Project: Configured Automation Environment & Project Structure Trainer-Guided Integrated Project: Automated Test Scripts (Individual Functional Scenarios) Trainer-Guided Integrated Project: Test Execution Reports and Status Trainer-Guided Integrated Project: Code Review and Feedback Session Artifacts	
PyTest	Introduction and installation of PyTest Naming conventions, test cases, and fixtures Running tests in various ways, skipping tests, grouping tests Parallel testing, parameterization, command line arguments HTML report generation Master automation testing with Pytest framework. Continuation of Integrated Project Deliverable: Agile and Functional Testing handson through Automation: • Trainer-Guided Integrated Project: Enhanced Automation Test Scripts • Trainer-Guided Integrated Project: Comprehensive Error Handling and Reporting • Trainer-Guided Integrated Project: Utility Class for Advanced Functions • Trainer-Guided Integrated Project: Code Review & Refactoring Artifacts	Day 41 to 43
GenAI Code Assist (GitHub Copilot): 3 Courses	Prompt Engineering & GitHub Copilot Usecase and Handbook Comprehensive GenAI Hands-On for Automation Developers	
Logs & Page Object Model	Generating logs Page Object Model pattern and Page Factory	Day 44
Hybrid Framework with POM – Live Project	Understanding Automation Frameworks and their types Designing a Hybrid Driven Framework step by step Project implementation with page objects, utilities, configuration files, automation, data-driven, and database test cases PyTest HTML reports, logs, email notifications Execution through command line and bat files	Day 45 to 47
Continuous Integration (Git, Github & Jenkins)	Introduction to Continuous Integration Workflow of Git & Github, Git commands Overview of Jenkins and Maven integration Running Selenium test scripts through Jenkins Continuation of Integrated Project Deliverable: Agile and Functional Testing handson through Automation • Trainer-Guided Integrated Project: The "Sprint Automation" Hybrid Framework Codebase • Trainer-Guided Integrated Project: Complete Automated Regression/Sprint Suite • Trainer-Guided Integrated Project: Enterprise-Level Reporting Suite (Extent & Allure) • Trainer-Guided Integrated Project: CI/CD Ready Execution Artifacts • Trainer-Guided Integrated Project: Finalized Agile Documentation & Handover Package	Day 48 to 50
Test Automation Frameworks(Robot Framework)	Continuation of Integrated Project Deliverable: Agile and Functional Testing handson through Automation Trainer-Guided Integrated Project: Refactored Automation Framework (Page Object Model Implementation) • Trainer-Guided Integrated Project: Data Management Strategy & Implementation (Apache POI) • Trainer-Guided Integrated Project: Execution Confirmation and Custom Logging Artifacts • Trainer-Guided Integrated Project: Updated Agile Artifacts & Traceability (Living Documentation) • Trainer-Guided Integrated Project: Robust Test Data Management via Test Automation Framework • Trainer-Guided Integrated Project: Comprehensive Execution Control • Trainer-Guided Integrated Project: Advanced Reporting with Extent Reports & Listeners • Trainer-Guided Integrated Project: Implementation of Assertions (Hard and Soft)	
BDD using Behave	BDD versus Traditional Development What is Behave? Install Behave and create Features Files and Step Definitions Using hooks Parameterizing the steps Data Driven Testing using Scenario Outline Tags Create Cucumber BDD testing framework using Python, Behave Read and write Cucumber BDD style tests in Python Cucumber BDD Framework supporting both Frontend with BDD Selenium and Backend with pure BDD Python Tips and tricks of finding elements, debugging issues, shortcuts with IDE, test methodologies, setup py file	Day 52 to 53
Behave BDD Hybrid Framework with PageObjectModel	Creating Feature files and Step Definitions Implemented Selenium Python code for all Scenarios Implementing hooks and utilities Implementing Page Object Model Optimizing Page Object Model Implementing Data Driven Testing No need for reading data from excel files and instead getting data from Generating Allure Reports Git, GitHub and Jenkins	Day 54 to 55
GenAI Code Assist (GitHub Copilot)	Comprehensive GenAI Hands-On for Automation Developers • Trainer-Guided Integrated Project: Chat Endpoint Basics • Trainer-Guided Integrated Project: Understanding Tokenization & Context Limits(Estimate & Optimize Tokens) • Trainer-Guided Integrated Project: Prompt Optimization via /ai/chat/with-usage • Trainer-Guided Integrated Project: Generate Edge Cases • Trainer-Guided Integrated Project: From User Story to Feature File (AI-Driven BDD Generation) • Trainer-Guided Integrated Project: Generate Locators for SauceDemo Login (Dynamic Locator Generation) • Trainer-Guided Integrated Project: AI-Augmented Framework • Trainer-Guided Integrated Project: Prompt Engineering Game	Day 56 to 57

Selenium Automation WebDriver



[Learn Selenium with Python, PyTest & Frameworks](#)

Refer all section in this Udemy course and complete the corresponding Subtopics.

2. Continuation of Integrated Project Deliverable: Agile and Functional Testing handson through Automation

Mandatory Hands-on

- Trainer-Guided Integrated Project: Automation Feasibility Report & Strategy
- Trainer-Guided Integrated Project: Configured Automation Environment & Project Structure
- Trainer-Guided Integrated Project: Automated Test Scripts (Individual Functional Scenarios)
- Trainer-Guided Integrated Project: Test Execution Reports and Status
- Trainer-Guided Integrated Project: Code Review and Feedback Session Artifacts

Handson

Selenium WebDriver Fundamentals & Project Kick-off

Please click this link to download Handson related to topics to be learnt as part of this milestone. Once practiced handson , please upload the solution file and upload here for evaluation by trainer.

GenAI Code Assist (GitHub Copilot)

Learning and practicing the below provided GenAI topics will equip you with essential, modern skills in QA Automation, preparing you for an evolving tech landscape

- **GenAI Foundation**

This section provides the foundational knowledge of how GenAI works and the critical importance of using it ethically and responsibly.

- Fundamentals of Generative AI [101-Basics]

- Generative AI Quick Assessment for Elearning Quiz [101-Basics]
- Introduction to Responsible AI
- Responsible AI-Deep Dive [101- BASICS]
- **Prompt Engineering:** The ability to communicate effectively with AI models is a core competency for maximizing their utility. This skill directly translates to generating higher quality code suggestions, more relevant test cases, and accurate documentation, saving significant time and effort.
 - Prompt Engineering Foundation - PY0009562
- **GitHub Copilot:** This section focuses on integrating AI assistance directly into the daily workflow. Learning to use GitHub Copilot effectively helps freshers write code faster, reduce bugs through AI-driven suggestions, and understand best practices through examples provided by the AI.

Introduction to GitHub Copilot

[GitHub Copilot for Beginners: AI Coding Crash Course](#)



Go through entire course. Implement the examples along with the author.

[GitHub Copilot - The Complete Guide – 2025](#)



Go through entire course. Implement the examples along with the author.

Usecase and Handbook & Comprehensive GenAI Hands-On for Automation

Developers: These exercises provide practical experience in optimizing AI outputs, managing token limitations, and generating functional, high-quality automation code, making the learning directly applicable to their job roles.

Day 41- 43

PyTest using GenAI Code Assist



[Learn Selenium with Python, PyTest & Frameworks](#)

Refer all section in this Udemy course and complete the corresponding Subtopics.

2. Continuation of Integrated Project Deliverable: Agile and Functional Testing handson through Automation

Mandatory Hands-on

- Trainer-Guided Integrated Project: Enhanced Automation Test Scripts
- Trainer-Guided Integrated Project: Comprehensive Error Handling and Reporting
- Trainer-Guided Integrated Project: Utility Class for Advanced Functions
- Trainer-Guided Integrated Project: Code Review & Refactoring Artifacts

Day 44- 47

Hybrid Framework with POM



[Learn Selenium with Python, PyTest & Frameworks](#)

Refer all section in this Udemy course and complete the corresponding Subtopics.

2. Continuation of Integrated Project Deliverable: Agile and Functional Testing handson through Automation

Mandatory Hands-on

- Trainer-Guided Integrated Project: Refactored Automation Framework (Page Object Model Implementation)
- Trainer-Guided Integrated Project: Data Management Strategy & Implementation (Apache POI)
- Trainer-Guided Integrated Project: Execution Confirmation and Custom Logging Artifacts
- Trainer-Guided Integrated Project: Updated Agile Artifacts & Traceability (Living Documentation)

3. Handson

Day 41-55 Frameworks, Patterns, and CICD

Please click this link to download Handson related to topics to be learnt as part of this milestone. Once practiced handson , please upload the solution file and upload here for evaluation by trainer.

Day 48 - 50

Continuous Integration (Git, Github & Jenkins)



Refer all section in this Udemy course and complete the corresponding Subtopics.

2. Continuation of Integrated Project Deliverable: Agile and Functional Testing handson through Automation

Mandatory Hands-on

- Trainer-Guided Integrated Project: The "Sprint Automation" Hybrid Framework Codebase
- Trainer-Guided Integrated Project: Complete Automated Regression/Sprint Suite
- Trainer-Guided Integrated Project: Enterprise-Level Reporting Suite (Extent & Allure)
- Trainer-Guided Integrated Project: CI/CD Ready Execution Artifacts
- Trainer-Guided Integrated Project: Finalized Agile Documentation & Handover Package

2. Continuation of Integrated Project Deliverable: Agile and Functional Testing handson through Automation

Mandatory Hands-on

- Trainer-Guided Integrated Project: Scalable Execution Infrastructure (Selenium Grid & Docker)
- Trainer-Guided Integrated Project: Behavior-Driven Development (BDD) Framework Integration
- Trainer-Guided Integrated Project: Continuous Integration & Delivery (CI/CD) Pipeline Integration

Day 51

Test Automation Frameworks(Robot Framework)

- Contents will be delivered by BU . This activity will be supported by Trainer . After the trainer covering the topics, trainees must prepare a document or PPT that includes their understanding and key points covered from the session provided by the trainer and upload here for evaluation by trainer.

Day 52 - 55

BDD using Behave



[Learn Selenium with Python, PyTest & Frameworks](#)

Refer all section in this Udemy course and complete the corresponding Subtopics.

2. Continuation of Integrated Project Deliverable: Agile and Functional Testing handson through Automation

Mandatory Hands-on

- Trainer-Guided Integrated Project: The "Sprint Automation" Hybrid Framework Codebase
- Trainer-Guided Integrated Project: Complete Automated Regression/Sprint Suite
- Trainer-Guided Integrated Project: Enterprise-Level Reporting Suite (Extent & Allure)
- Trainer-Guided Integrated Project: CI/CD Ready Execution Artifacts
- Trainer-Guided Integrated Project: Finalized Agile Documentation & Handover Package.

Project – Hackathon

As part of Final evaluation, Trainees must do below Hackathon projects has been shared below and demonstrate these projects in the Final evaluation to the evaluator.

Docker



[Learn Selenium with Java, Cucumber & Frameworks](#)

Refer all section in this Udemy course and complete the corresponding Subtopics.

2. Continuation of Integrated Project Deliverable: Agile and Functional Testing handson through Automation

Mandatory Hands-on

- Setup Grid and parallel execution

Additional Hands-on

- GIT installation & commands implementation
- Install Jenkins and Creation of new job

- Mavenization using Eclipse M2E plugin Options and CLI commands

3. QEA Role specific Automation Technical Handson

[Click here to download the BU Aligned Technical Training Contents](#)

Chapter 17_Handson

Chapter 17_Video Learning Ref.Content_Part 1

Chapter 17_Video Learning Ref.Content_Part 2

Chapter 17_Video Learning Ref.Content_Part 3

Chapter 17_Video Learning Ref.Content_Part 4

Chapter 17_Video Learning Ref.Content_Part 5

Chapter 18_Handson

Chapter 18_PPT Learning Ref.Content

Chapter 19_Handson

Chapter19_PPT Learning Ref.Content

Chapter 21_Handson

Chapter 21_PPT Learning Ref.Content

Chapter 22_Handson

Chapter 23_Handson

Chapter 24_Handson

BDD and Cucumber



[Learn Selenium with Java, Cucumber & Frameworks](#)

Refer all section in this Udemy course and complete the corresponding Subtopics.

- Simple cucumber project with phptravels
- Simple cucumber project using datatable
- Simple cucumber project using scenario outline

5. Continuation of Integrated Project Deliverable: Agile and Functional Testing handson through Automation

- Trainer-Guided Integrated Project: Scalable Execution Infrastructure (Selenium Grid & Docker)
- Trainer-Guided Integrated Project: Behavior-Driven Development (BDD) Framework Integration
- Trainer-Guided Integrated Project: Continuous Integration & Delivery (CI/CD) Pipeline Integration

Quiz
Devops

Day 56 - 57

STAGE: Modern AI-Assisted Automation

Comprehensive GenAI Hands-On for Automation Developers

- Trainer-Guided Integrated Project: Chat Endpoint Basics
- Trainer-Guided Integrated Project: Understanding Tokenization & Context Limits(Estimate & Optimize Tokens)
- Trainer-Guided Integrated Project: Prompt Optimization via /ai/chat/with-usage
- Trainer-Guided Integrated Project: Generate Edge Cases
- Trainer-Guided Integrated Project: From User Story to Feature File (AI-Driven BDD Generation)
- Trainer-Guided Integrated Project: Generate Locators for SauceDemo Login (Dynamic Locator Generation)
- Trainer-Guided Integrated Project: AI-Augmented Framework
- Trainer-Guided Integrated Project: Prompt Engineering Game

Project – Hackathon

Deliver your project and get the review comments from SME.

The deliverables of the Hackathon will be evaluated by the BU SME.
Project Evaluation will be based on:

- Source Code
- Functionality Completion, Usage of Features, Code Quality
- Demo of Output

Day 58 - 60

Evaluation:
Final Project Evaluation + Final Technical Evaluation

The overall Final Technical evaluation will cover topics specified in the curriculum and will be conducted through a video interview on the GenC Learn Platform.

Trainees are given two attempts to pass the final evaluation. If a Trainee receives a RAG status of “RED” or “AMBER” status on their first attempt, they have one opportunity to improve and achieve a “GREEN” status on their second attempt.

What is Final Evaluation?

The Final Evaluation will be conducted to certify whether a GenC is eligible to enter into the BU or not. The skill of a GenC will be gauged on the application development and overall technical knowhow towards the end of GenC Training.

Tech SME from BU will be conducting the final tech evaluation. As a fallback, the project mentor can also steer this activity.

The final evaluation will be conducted as two phases. They are the following

- 1. Final Technical Evaluation**
- 2. Final Project Evaluation**

The mode of these evaluations will be any one of the following:

- F2F(face to face)
- Video Based

1. Final Technical Evaluation (FTE)

The BU Mentor will interview the GenC on various skills achieved throughout the training program and put a score which will be considered for the final PHS of the GenC.

2. Final Project Evaluation (FPE)

In this evaluation, the BU Mentor will be verifying the skills of a GenC on a project perspective. End of this evaluation, the BU Mentor will score the GenC's work based on various evaluation criterions.



How to learn each day?

Each day has a set of learning objectives. These learning objectives can be met by going through the Udemy courses and by completing the hands-on exercises mentioned in the daily plan.

The below strategies will help you decide the learning approach.

Learning Strategy & Approach

Find below few imaginary profiles. For each of these profiles we have defined a recommended learning approach. This is not an exhaustive list. The approaches below might help invent a new way of learning.

Profile #1



Harry Reacher

Engineering Discipline: Electronics

Skills: Python, Ruby on Rails, nginx

Project: Mining Crime Data to get Route Cause Insights

Learning Approach to Programming Languages: I do not want to waste my time learning. I am more practice oriented. I want to work on the problem immediately

What will work for me?

- Directly complete hands on exercises
- Refer Internet or Udemy Courses
- If hands on are implemented early, clarify your friends questions and troubleshoot their issues

Profile #2



Olivia Richards

Engineering Discipline: Computer Science

Skills: Java, C, C++

Project: Library Management System

Learning Approach to Programming Languages: I have interest, but I don't know where to start.

What will work for me?

- Go through the recommended Udemy Course
- Try completing the hands on exercises
- Get your clarifications solved with help from Tech SME
- Get help from other learners in your batch whom had already completed

Profile #3

Greg Anderson

Engineering Discipline: Civil



Skills: C

Project: Fiber reinforced concrete

Learning Approach to Programming Languages: I am scared of programming languages. I haven't got my hands dirty with coding

What will work for me?

- Go through the recommended Udemy Course
- Implement the coding along with the author of the Udemy Course
- Try completing the hands on exercises
- Clarify queries with SME
- Troubleshoot programming issues with help from SME or learner from your classroom whom had already completed

FAQ

1. Who can participate in this program?

Students who have enrolled for Full Internship Program (or) the Cognizant on-boarded GEN Cs can participate in this program.

2. Is there any pre-learning I should do?

No. This program is open to all students from any academic discipline.

3. How will I know my RAG status?

It will be shown to you in the GEN C learn Platform, in your Home Page.

4. Whom do I reach out in case of any queries?

Coach is your point of contact.

5. What is the significance of Hands-on in the overall learning journey?

Hands-on focuses on specific topics in a Skill, which you can try and execute in the Platform. Group of such Hands-on exercises will be packaged together as a Code Challenge. This Code Challenge will allow you to benchmark your skills in the learning journey. Hands On/ Code Challenges/ ICT are learning components which will help you in understanding the skills better.

6. What is Code Challenge?

A problem statement will be provided to you and you need to solve it using a single skill.

7. What is Integrated Capability Test (ICT)?
A case study problem statement will be provided to you, that you may need solve using the combination of Skills learnt in the given stage.
8. How many attempts are provided for the Coding challenge and ICTs? Is it open all the time for practice?
The Coding challenges and ICTs are open and there are 3 attempts to take them up.
9. What is the entry criteria for qualifier?
A minimum of 70% hands-on completion and attempt in the CC & ICT is the eligibility criteria for qualifier.
10. What skills are covered in the qualifier?
The skills of Stage 1 are covered in the qualifier. Only ONE attempt is provided to clear with a minimum score of 70%
11. What if I fail in the Interim evaluation?
Your coach will notify your performance in the Interim evaluation. However you can continue with the learning.
12. How many chances will I get in the Final evaluation?
You'll get 2 chances in the Final evaluation which covers ALL the skills in the learning journey.
13. Whom do I reach out in case of any queries?
Coach is your point of contact.